

UNIVERSITY OF CLAUDE · FREE RESOURCE MAP

Resource Map

CSCE 671 Human-Computer Interaction — module-by-module mapping onto free university courseware

HOW TO USE THIS

This is your lecture hall. For each module: watch or read the mapped external material first, then work the module slides and notes, which are written to assume you have seen it.

Where a specific lecture number is given, that is the one that covers this module. Where a book chapter is given, read that chapter, not the whole book.

Everything here is free. Resources remain under their own licences — link to them, do not redistribute them.

Primary sources for the whole course

Resource	Role in this course
Norman — <i>The Design of Everyday Things</i> , revised edition	The primary text for Modules 01, 04, and 05. The affordance, signifier, and conceptual model vocabulary this course uses comes from here, and the examples are still the best ones. Library copy.
Johnson — <i>Designing with the Mind in Mind</i> , 3rd edition	Modules 02 and 03. The perceptual and cognitive research, written for designers and honest about effect sizes — which is unusual in this area. Library copy.
Krug — <i>Don't Make Me Think</i> , revisited	Modules 05 and 07. Short, practical, and its discount usability testing chapter is the best introduction to Module 07 there is.
Nielsen Norman Group articles, and the heuristics (free)	Modules 05 and 07's working reference — the heuristics, the discount methods, and a large body of reported studies. Read the method sections, not only the conclusions.
Lazar, Feng & Hochheiser — <i>Research Methods in Human-Computer Interaction</i> , 2nd edition	Modules 08 and 09. The methods treated properly, including the ethics and the statistics. Library copy.
The CHI proceedings (free via the ACM OA programme)	The field's primary literature. Module 13's exercises use it; read the limitations sections, which is where the honest claims are.

Module-by-module map

Module 01 — Why Interfaces Fail

Resource	What it covers for this module
Norman — The Design of Everyday Things, chapters 1 and 2	§1, §3, and §4 — the gulfs, the affordance/signifier distinction, and the human-error argument, from the source.
Nielsen — the usability heuristics, and the articles on scanning (free)	§2's non-reading finding and §3's signifier problems , with the studies behind them.
Norman — Affordances and Design (free)	§3's distinction, clarified by its author — written specifically because the term had been misused.
Woods & Cook — on human error in complex systems	§4's argument from safety engineering — the intellectual source of the blameless framing. Library copy.

Module 02 — Perception

Resource	What it covers for this module
Johnson — Designing with the Mind in Mind, chapters 1 through 5	The whole module , with the research cited and the effect sizes given. Library copy.
Ware — Information Visualization: Perception for Design	§2 and §3 in depth — the standard reference, and CSCE 679's primary text as well. Library copy.
Healey — Perception in Visualization (free)	§3's preattentive features, with interactive demonstrations — the conjunction search effect is worth experiencing rather than reading about.
Simons & Levin — Change blindness (free)	§4 in the original — and the magnitude of the effect is the part that changes how you design feedback.

Module 03 — Attention, Memory, and Error

Resource	What it covers for this module
Johnson — Designing with the Mind in Mind, chapters 7 through 11	§1 through §3 , with the corrected memory figures and the sources. Library copy.
Cowan — The magical number 4 in short-term memory (free)	§2's correction in the original — why the figure is four rather than seven, and what a chunk is.
Norman — The Design of Everyday Things, chapter 5	§4's slip and mistake taxonomy , with the sub-categories and the fixes.
Mark, Gudith & Klocke — The cost of interrupted work (free)	§1's switching cost, measured — and the finding that people compensate by working faster, at a cost in stress.

Module 04 — Mental Models

Resource	What it covers for this module
Norman — The Design of Everyday Things, chapter 3	The whole module — the three models, the system image, and the knowledge-in-the-world argument.
Johnson-Laird — Mental Models	§1 and §2's cognitive basis — where the term comes from and what it is claimed to be. Library copy.
Kuang & Fabricant — User Friendly	§2's analogies, historically — how particular metaphors were chosen and what they cost later. Library copy.
Nielsen Norman Group on mental models (free)	§4's elicitation, practically — with the interview protocols.

Module 05 — Interaction Design Principles

Resource	What it covers for this module
Nielsen — Ten Usability Heuristics (free)	§1's list in its canonical form , and the one most reviews are run against.
Norman — The Design of Everyday Things, chapters 4 and 7	§1's constraints and §2's conflicts , with the mechanisms developed properly.
Nielsen & Molich — Heuristic evaluation of user interfaces (free)	§4 in the original — including the measured overlap between reviewers, which is the key number.
Thaler & Sunstein, and the default-effect literature	§3's effect outside software — where the default-dominance finding was established.

Module 06 — Prototyping

Resource	What it covers for this module
Buxton — Sketching User Experiences	§1's argument at length — why sketching and prototyping are different activities. Library copy.
Snyder — Paper Prototyping	§1 and §2 practically — how to run a paper prototype session, in detail. Library copy.
Dahlbäck, Jönsson & Ahrenberg — Wizard of Oz studies (free)	§3's method and its pitfalls — including the wizard-is-too-clever problem.
Nielsen Norman Group on prototyping fidelity (free)	§2's decision, practically — and the independent dimensions made explicit.

Module 07 — Usability Testing

Resource	What it covers for this module
Krug — Rocket Surgery Made Easy	The whole module, practically — the cheapest possible version of the method, argued for convincingly. Library copy.

Resource	What it covers for this module
Nielsen — Why You Only Need to Test with 5 Users (free)	§2's argument in its canonical form — read it with the critiques, which are also linked there.
Dumas & Redish — A Practical Guide to Usability Testing	§1, §3, and §4 thoroughly — the task writing and the facilitation in detail. Library copy.
Nielsen Norman Group on think-aloud protocols (free)	§1's protocol , with the facilitator scripts and the common mistakes.

Module 08 — Controlled Experiments with People

Resource	What it covers for this module
Lazar, Feng & Hochheiser — Research Methods in HCI, chapters 2 through 5	The whole module — the designs, the statistics, and the ethics, treated properly. Library copy.
Cairns — Doing Better Statistics in Human-Computer Interaction	§3 specifically — and honest about how badly the field has historically done this. Library copy.
Dragicevic — Fair Statistical Communication in HCI (free)	§3's reporting recommendations — effect sizes and intervals instead of significance, argued for this field.
The Belmont Report, and your institution's ethics guidance (free)	§4's obligations in their original framing — and whatever local process applies to you.

Module 09 — Qualitative Methods

Resource	What it covers for this module
Beyer & Holtzblatt — Contextual Design	§3's method in full — the apprentice stance and the artefact collection, from its originators. Library copy.
Portigal — Interviewing Users	§2 practically — short, and the question-phrasing advice is the best available. Library copy.
Lazar, Feng & Hochheiser, the qualitative chapters	§4's analysis — coding, agreement, and what the results support. Library copy.
Nisbett & Wilson — Telling more than we can know (free)	§3's finding in the original — the classic result on the unreliability of introspective reports.

Module 10 — Interfaces for Programmers

Resource	What it covers for this module
Myers & Stylos — Improving API usability (free)	§1 with the empirical studies — programmers watched using APIs, and what the failures were.

Resource	What it covers for this module
Procházka — Diátaxis documentation framework (free)	§3's four kinds — the clearest statement of the distinction and why conflating them fails.
Rust's error message guidelines, and the Elm compiler's approach (free)	§2 done well, deliberately — two projects that treated error messages as an interface and documented how.
Bloch — How to Design a Good API and Why it Matters (free talk)	§1 from a practitioner — and much of it maps onto §1's table without being framed that way.

Module 11 — Input and Output

Resource	What it covers for this module
MacKenzie — Fitts' law as a research and design tool (free)	§1 in depth — the formulations, the measurement method, and the two-dimensional extensions.
Card, Moran & Newell — The Psychology of Human-Computer Interaction	§1 and §3's foundations — where the timing constants in this module come from. Library copy.
Nielsen — Response Times: The 3 Important Limits (free)	§3's thresholds , with the sources and the behavioural consequences.
Material and Apple human interface guidelines (free)	§2's touch constraints as platform guidance — including the minimum target sizes and the reasons.

Module 12 — Manipulative Design

Resource	What it covers for this module
Brignull — Deceptive Patterns (free)	§1's catalogue — the patterns named, with examples and the enforcement history.
Mathur et al. — Dark Patterns at Scale (free)	§1 measured — an automated survey of eleven thousand shopping sites, with prevalence figures.
Thaler & Sunstein — Nudge, and its critics	§3's difficult cases — the case for benign paternalism, and the objections to it. Library copy.
Susser, Roessler & Nissenbaum — Online Manipulation (free)	§3's distinction, argued properly — manipulation as bypassing rather than engaging deliberation.

Module 13 — Claiming Usability Honestly

Resource	What it covers for this module
Lazar, Feng & Hochheiser, the reporting chapters	§1 and §2 — what each method supports, and how to report it. Library copy.

Resource	What it covers for this module
Cockton — on the limits of usability claims (free via the CHI proceedings)	§2's overclaims, in the field's own literature — and the limitations sections of CHI papers are the model to follow.
Greenberg & Buxton — Usability Evaluation Considered Harmful (free)	§1's first two rows argued sharply — when evaluation is the wrong activity, which is a useful corrective.
Dragicevic — Fair Statistical Communication in HCI (free)	§2's third claim form — intervals rather than significance, and why.